

Input Parameters

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

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1 Summary

This report provides an overview of all input parameters used in the METIMMOX-1 health economic model, organized into global model settings and variable parameters across base case, DSA, and PSA analyses.

2 Global Model Parameters

These parameters define the overall model structure and are held constant across all analyses.

Table 1: Global Model Parameters

Parameter	Value	Description	Source
Time Horizon	520 weeks (10 years)	Model time horizon	Clinical follow-up period
Cycle Length	1 week	Cycle length for discrete-time model	Model assumption
Discount Rate	4% per annum	Annual discount rate for costs and effects	Model specification
WTP Threshold	€51,000 per QALY	Willingness-to-pay threshold	Norwegian threshold based on severity (14.5 QALYs lost for diagnosis, based on a health 65-year old)
PSA Sample Size	5,000 iterations	Number of Monte Carlo iterations for PSA	Computational adequacy
Sampling Models	5,000 models	Resampled models for survival uncertainty	Statistical adequacy
DSA Variation Range	$\pm 20\%$	Parameter variation range for DSA	Standard practice
Model Structure	0 (joint)	Survival model formula: 0=joint (all biomarkers + all interactions), 1=focused (per-strategy formulas: CRP/TMB_BRAF with both interactions, TLR with TLR-only interaction), 2=separate (one biomarker only)	Methodological choice

3 Model Input Parameters

This table presents all variable model parameters organized by type, showing their base case values, DSA variation ranges, and PSA distribution specifications.

Table 2: Model Input Parameters by Type

Type	Parameter	Base Case	DSA Range	PSA Distribution
Cost Parameters				
Cost	Nivolumab per administration	€13,923	€11,138 - €16,708	Gamma (shape=25, rate=0.0018)
Cost	FLOX per administration	€427	€342 - €512	Gamma (shape=25, rate=0.0586)
Cost	CT scan	€386	€309 - €463	Gamma (shape=25, rate=0.0648)
Cost	Blood tests	€0	Not varied	Fixed at 0
Cost	NGS testing	€1,439	€1,151 - €1,727	Gamma (shape=25, rate=0.0174)
Cost	Standard outpatient visit	€33	Not varied	Gamma (shape=25, rate=0.7576)
Cost	Baseline visit	€530	Not varied	Gamma (shape=25, rate=0.0472)
Cost	Follow-up visit	€33	Not varied	Gamma (shape=25, rate=0.7576)
Cost	End-of-life care	€13,803	€11,042 - €16,564	Gamma (shape=25, rate=0.0018)
Utility Parameters				
Utility	Non-progressed state	0.9077	0.7262 - 1.0000	Beta (=41.0, =4.2)
Utility	Progressed state	0.9005	0.7204 - 1.0000	Beta (=40.6, =4.5)
Prevalence Parameters				
Prevalence	CRP biomarker positive	30%	24% - 36%	Beta (CV=15%)
Prevalence	TLR biomarker positive	25%	20% - 30%	Beta (CV=15%)
Prevalence	TMB/BRAF biomarker positive	20%	16% - 24%	Beta (CV=15%)
Survival Parameters				
Survival	OS and PFS curves by biomarker status	Fitted parametric survival models	Not varied	Survival resampling (5,000 models)

Notes:

DSA ranges calculated as base case $\pm$ 20%. Utility upper bounds capped at 1.0.

PSA: Gamma distributions for costs (CV=20%), Beta for utilities (CV=15%).

Survival curves use non-parametric resampling with age and sex adjustments.

4 PSA Parameter Distributions

This section displays histograms of all sampled parameters from the probabilistic sensitivity analysis (PSA), showing the distribution shapes and ranges used in the Monte Carlo simulation.

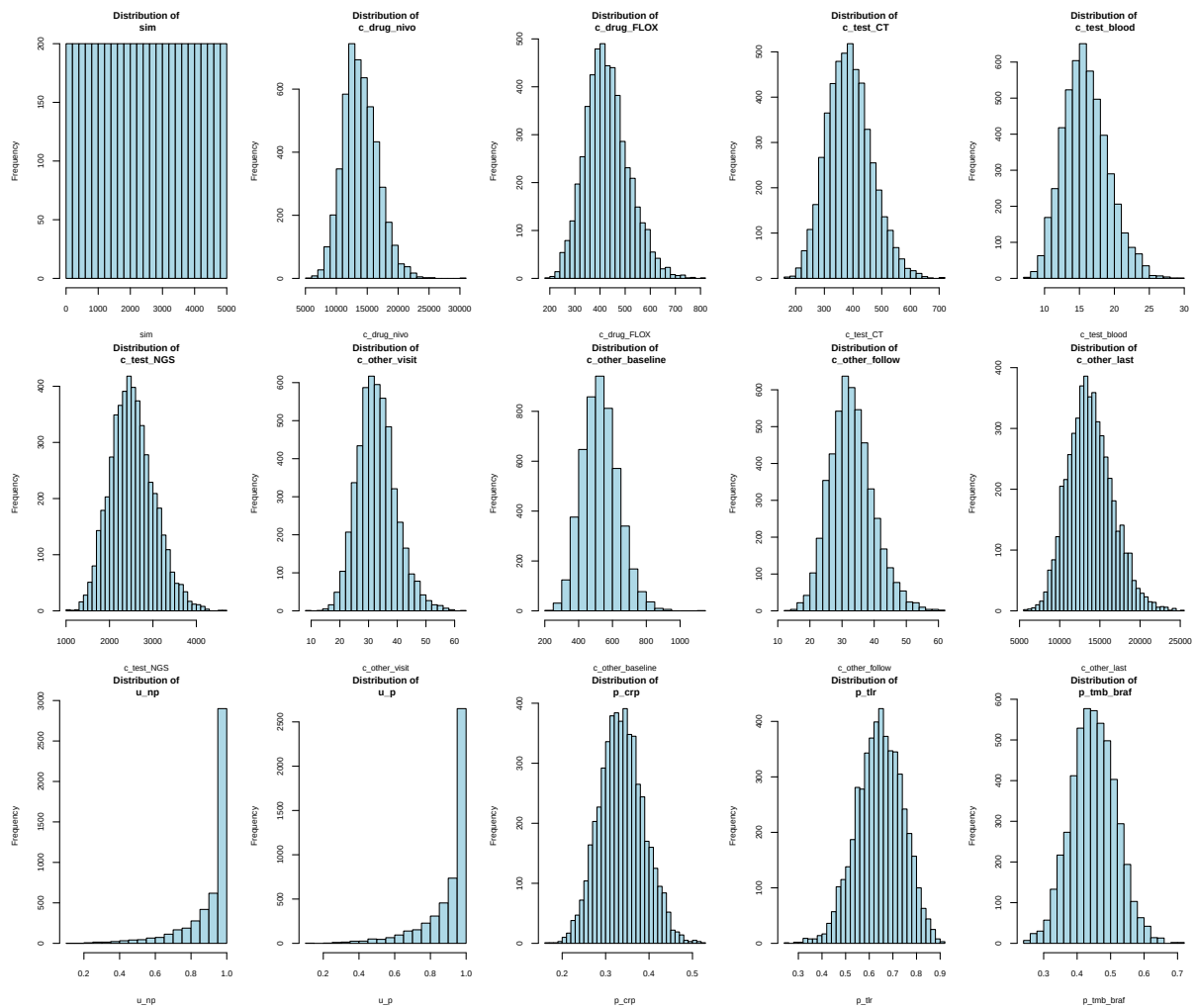


Figure 1: Distribution of PSA Parameters

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